

PART 3

Business Strategy Recommendation

The developed model attained an 100% fraud precision, which means that all transactions it identified as fraudulent were actually fraudulent. However, the model identified only 68% (136) of frauds, while the remaining 64 fraudulent transactions went undetected in the test set.

To balance customer experience and fraud, I recommend a three-tiered approach rather than blocking all transactions that are predicted to be fraudulent:

1. **Low-Risk Transactions:** Any transaction with a low fraud score should be approved automatically. This will reduce friction for genuine customers and aid in transaction completion rates.
2. **Medium-Risk Transactions:** Transactions with moderate fraud scores should be channelled for further authentication such as OTP, device verification, or step-up authentication. This will reduce the likelihood of blocking legitimate users while providing an additional layer of protection.
3. **High-Risk Transactions:** Transactions with high fraud scores should be automatically declined or held for manual review. Given the model's very high precision, transactions classified as high risk are highly likely to be fraudulent.

Through this approach, the organization can mitigate fraud losses, compliance risks, and ensure that legitimate high-value transactions are not rejected. The strategy also allows for business teams to adapt their approval, review and decline thresholds according to evolving fraud patterns and risk appetites without changing the underlying model.